

Amanda Juvera

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EDUCATION

University of Michigan

Grad. May 2027

Bachelor of Science in Computer Science

Ann Arbor, MI

- **Relevant Coursework:** Distributed Systems, Operating Systems, Advanced Operating Systems, Data Structures and Algorithms, Machine Learning, Computer Organization, Object Orientated Programming
- **Awards:** Blue Ribbon Undergraduate Research Award, University Honors, LSA Scholarship Recipient

EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Oracle

Santa Clara, CA

- Redesigned SBOM generation in Syft (open-source Go library) to preserve package ownership context across teams, eliminating 100% of false-positive vulnerability audits for SmartNIC artifacts and surfacing 1,000+ undetected vulnerabilities.
- Shipped a custom build service configuration that scaled security scanning from a single team to every Oracle Cloud Infrastructure (OCI) artifact, extending coverage org-wide with no added operator workload.

Software Engineer

Jan 2026 – May 2026

Bananza Marketing Agency

Prague, Czech Republic

- Shipped a nightly Python pipeline consolidating performance metrics for 50+ clients into a centralized workspace, cutting 5+ hours per week of manual data entry.
- Modeled client records as structured databases over paginated REST APIs with schema mapping and rate-limit handling, letting non-technical strategists build client-facing reports without engineering support.
- Deployed on GitHub Actions with encrypted secrets and dual cron schedules, eliminating all manual runs.

Software Engineer Intern

May 2025 – Aug 2025

Retrospect

Ann Arbor, MI

- Built a computer vision pipeline using HSV filtering and region-of-interest cropping to classify traffic light state from low-quality video at 90% accuracy, reclaiming 10+ hours per week for safety reviewers.
- Integrated traffic light detection end-to-end into RiskEngine, a production tool evaluating autonomous vehicle safety across large volumes of recorded drive data.

Computer Science Mentor

Aug 2024 – Present

University of Michigan, Computer Science and Engineering

Ann Arbor, MI

- Mentored students across four semesters in data structures, memory management, concurrency, and C++ debugging, giving and receiving rapid feedback under deadline.

Research Assistant

Oct 2023 – Apr 2024

University of Michigan, Maldonado Lab

Ann Arbor, MI

- Built LabVIEW automation software for Scanning Electrochemical Probe Microscopy, cutting experiment runtime by 50% and removing manual instrument operation from the research workflow.

PROJECTS

Personal Website & CMS | TypeScript, Next.js, React, Node.js, PostgreSQL, Prisma, Zod, Vercel

- Built and deployed a full-stack portfolio and CMS in Next.js 16 / React 19 with PostgreSQL + Prisma, an OAuth-gated admin dashboard, and a Canvas-animated UI that models the site as a browsable filesystem.
- Hardened the public API with Zod schema validation and salted-hash IP rate limiting, and automated a nightly cron sync of GitHub REST API repo metadata.

Navi, Road Trip Navigation Application (in progress) | Swift, JavaScript, Python, MapKit, Cloudflare Workers

- Building an “on the way” stop search that ranks candidates by added travel time along the route corridor, reducing wasted detours compared to radius-based search.
- Designed a refueling scheduler over tank capacity, fuel efficiency, and crowd-sourced gas prices, minimizing total trip fuel cost with selectable strategies for fewest stops versus lowest spend.

Operating Systems Components | C++, POSIX, TCP

- **Multi-threaded Network File Server:** Built a concurrent, crash-consistent file server over TCP sockets using fine-grained reader-writer locking, hand-over-hand traversal, and ordered disk writes.
- **Virtual Memory Manager:** Designed a demand-paged system with swap- and file-backed pages, copy-on-write `fork()`, and clock page replacement, minimizing disk I/O on page faults.
- **Thread Library:** Developed a kernel-level library with `ucontext` switching, mutexes, condition variables, and interrupt-safe synchronization, enabling preemptive multitasking across multiple CPUs.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, C, Java, Swift, SQL, Go

Frameworks and Libraries: React, Next.js, Node.js, Prisma, Zod, PostgreSQL

Tools and Infrastructure: Git, GitHub Actions, Linux, Vercel, REST APIs, AI coding tools (Claude Code, Codex)